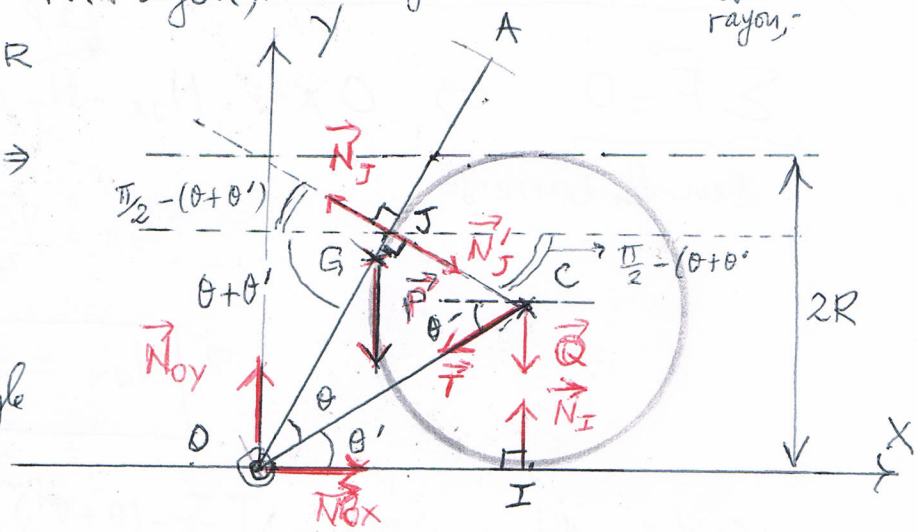
TD 2, EXO 4

R : rayon, * OI tangente au cercle en I $\Rightarrow OI \perp CI$

* OA tangente au cercle $\Rightarrow OA \perp CJ$ (rayon)

$OA = 3R$, $OC = 2R$



* Triangle $O\hat{C}J$ rectangle $\Rightarrow$

$\sin\theta = \frac{CJ}{OC} = \frac{R}{2R} = \frac{1}{2}$

$\Rightarrow \theta = 30°$

* Triangle OIC rectangle

$\Rightarrow \sin\theta' = \frac{CI}{OC} = \frac{R}{2R} = \frac{1}{2}$

$\Rightarrow \theta' = 30°$

* Toujours, triangle OCJ rectangle $\Rightarrow \cos 30° = \frac{OJ}{OC} \Rightarrow OJ = OC\cos 30° = 2R\frac{\sqrt{3}}{2}$

Donc $OJ = R\sqrt{3}$

La barre OA est en équilibre :

* Donc : $\sum \vec{M}_O = 0 \Rightarrow \vec{M}_O(\vec{N}_{ox}) + \vec{M}_O(\vec{N}_{oy}) + \vec{M}_O(\vec{P}) + \vec{M}_O(\vec{N}_J) = 0$ (0, 0, $\vec{j}$)

$\Rightarrow \overrightarrow{OG} \wedge \vec{P} + \overrightarrow{OJ} \wedge \vec{N}_J = 0$ (la barre)

$$\Rightarrow \begin{vmatrix} \frac{3R}{2}\cos(\theta+\theta') & \frac{3R}{2}\sin(\theta+\theta') & 0 \\ OG & 0 & 0 \\ 0 & -P & 0 \end{vmatrix} + \begin{vmatrix} OJ\cos(\theta+\theta') & OJ\sin(\theta+\theta') & 0 \\ -N_J\cos\left(\frac{\pi}{2}-(\theta+\theta')\right) & N_J\sin\left(\frac{\pi}{2}-(\theta+\theta')\right) & 0 \end{vmatrix} = 0$$

($\theta+\theta' = 60°$, $OG = \frac{3R}{2}$, $OJ = R\sqrt{3}$, $\frac{\pi}{2}-60° = 30°$)

$$\Rightarrow \begin{vmatrix} \frac{3R}{4} & \frac{3\sqrt{3}R}{4} & 0 \\ 0 & -P & 0 \end{vmatrix} + \begin{vmatrix} R\frac{\sqrt{3}}{2} & R\frac{(\sqrt{3})^2}{2} & 0 \\ -N_J\frac{\sqrt{3}}{2} & \frac{N_J}{2} & 0 \end{vmatrix} = 0$$

$$\Rightarrow -\frac{3RP}{4}\vec{k} + \vec{k}\left(\frac{RN_J\sqrt{3}}{4} + \frac{RN_J 3\sqrt{3}}{4}\right) = 0 \Rightarrow \frac{-3RP}{4} + RN_J\sqrt{3} = 0 \Rightarrow \boxed{N_J = \frac{\sqrt{3}}{4}P} = N'_J$$

* Le Disque est en équilibre :

* $\sum \vec{F} = \vec{0}$ (sur le Disque) $\Rightarrow$ OX : $-T\cos\theta' + N'_J\cos\left(\frac{\pi}{2}-(\theta+\theta')\right) = 0$ ($\theta' = 30°$, $N'_J = N_J$, $\frac{\pi}{2}-(\theta+\theta') = 30°$)

$\Rightarrow -T\frac{\sqrt{3}}{2} + \frac{\sqrt{3}P}{4}\frac{\sqrt{3}}{2} = 0 \Rightarrow \boxed{T = \frac{\sqrt{3}}{4}P}$

OY : $-T\sin\theta' - N'_J\sin\left[\frac{\pi}{2}-(\theta+\theta')\right] - Q + N_I = 0$ ($\theta' = 30°$, $N'_J = N_J$, $\frac{\pi}{2}-(\theta+\theta') = 30°$)

$-\frac{\sqrt{3}}{4}P\times\frac{1}{2} - \frac{\sqrt{3}}{4}P\times\frac{1}{2} - Q + N_I = 0 \Rightarrow \boxed{N_I = Q + \frac{\sqrt{3}}{4}P}$

Continuer page suivante →

* Revenons à la barre

$\underbrace{\sum \vec{F} = 0}_{\text{sur la barre}} \Rightarrow OX : \quad N_{ox} - N_J \cos\left[\frac{\pi}{2} - (\theta+\theta')\right] = 0$

($N_J = \frac{\sqrt{3}}{4} P$, $\frac{\pi}{2} - (\theta+\theta') = 30°$)

$\Rightarrow N_{ox} = \frac{\sqrt{3}}{4} P \frac{\sqrt{3}}{2} = \frac{3P}{8}$

$$\Rightarrow \boxed{N_{ox} = \frac{3P}{8}}$$

$OY : \quad N_{oy} + N_J \sin\left[\frac{\pi}{2} - (\theta+\theta')\right] - P = 0$

$\Rightarrow N_{oy} = -\frac{\sqrt{3}}{4} P \cdot \frac{1}{2} + P \Rightarrow \boxed{N_{oy} = P\left(1 - \frac{\sqrt{3}}{8}\right)}$

$$N_o = \sqrt{N_{ox}^2 + N_{oy}^2} = \sqrt{\frac{19 - 4\sqrt{3}}{16}}\, P$$

$$\boxed{N_o = \frac{\sqrt{19 - 4\sqrt{3}}}{4} P}$$